MANUEL PAEZ

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EDUCATION

Columbia University New York, NY

Master of Science, Computer Science, GPA: 3.83/4.0 Jan. 2025 - Expected Dec. 2026

Graduate Coursework: Advanced Probability Theory, Computational Learning Theory, Intro to Databases, Probability II, Projects in CS, Statistical Inference, Theoretical Foundations of LLMs.

Bachelor of Arts, Mathematics Sept. 2023 - May 2026

Bachelor of Arts, Computer Science Sept. 2019 - May 2023

Relevant Coursework: Advanced Algorithms, Advanced Topics in Deep Learning, Algorithms for Massive Data, Analysis and Probability I, Honors Complex Variables, Machine Learning Theory, Natural Artificial Neural Networks, Real Analysis I and II, Theoretical Neuroscience, Unsupervised Learning, Intro to Quantum Computing.

RESEARCH EXPERIENCE

Columbia University, Department of Computer Science New York, NY

Student Researcher; Advisor: Professor Alexandr Andoni Jan. 2024 - Present

Geometric Spanners and Approximate Nearest-Neighbor Search

- Established subquadratic-time constructions of Euclidean Steiner spanners for $c > 1$ and randomized spanner constructions in Hamming space with dimension $d = (1 + \varepsilon) \log n$.
- Developed randomized algorithms for constructing sparse geometric spanners via batched approximate nearest-neighbor search, achieving O(ck) distortion with $\tilde{O}(kn^{1+1/(2k)} \log \Delta)$ edges; established matching $\Omega(ck)$ distortion lower bounds under restricted black-box oracle access.
- Extended the framework to subquadratic approximation of Wasserstein-q distances using bounded-hop spanners and minimum-cost flow on sparse directed graphs.

Low-Dimensional ℓ_1 Embeddings of Ising Disagreement Metrics

- Investigated structural dimension reduction of disagreement metrics induced by Ising Gibbs measures, seeking constant-distortion ℓ_1 embeddings in polylogarithmic dimension beyond standard sampling-based constructions.
- Established quantitative correlation-decay bounds for bounded-degree Ising models under Dobrushin uniqueness, yielding $1 \pm \varepsilon$ approximations by conditional product metrics at locality radius $O(\eta^{-1}\log(1/(\varepsilon\eta)))$.
- Investigated Poincaré inequalities, spectral-gap bounds, and approximate tensorization to control correlations and develop low-dimensional ℓ_1 embedding guarantees for Gibbs disagreement metrics.

Differentially Private Approximate Nearest-Neighbor Counting

- Investigated packing-based lower bound arguments for differentially private approximate near-neighbor counting in high-dimensional space, analyzing privacy, approximation, and additive-error tradeoffs.

Flatiron Institute, Simons Foundation New York, NY

Neural Circuits and Algorithms Group, Center for Computational Neuroscience

Visiting Researcher; Advisor: Dmitri "Mitya" Chklovskii Feb. 2026 - Present

- Developed and implemented offline and online nonnegative asymmetric similarity matching (NASM) algorithms for learning local predictive directions in nonlinear dynamical systems. Code predominantly in Python.
- Evaluated predictive-direction recovery on stochastic saddle-point, generalized Lotka-Volterra, and spherical dynamics, achieving approximately 99.9% precision on the first two systems, respectively.
- Benchmarked NASM using matched-filter cosine similarity, normalized mutual information, and predictive information; online learning reached a 10^{-3} relative parameter-change threshold within 42,000–58,000 samples on generalized Lotka–Volterra dynamics.

Visiting Scholar Fellow; Advisor: Dmitri "Mitya" Chklovskii Nov. 2024 - Aug. 2025

- Investigated biologically plausible, self-supervised representation learning through Rectified Spectral Units (ReSUs) and past–future canonical correlation analysis (CCA) for extracting low-dimensional predictive features.

- Implemented and evaluated a multilayer ReSU architecture, analyzing learned temporal filters, rectified representations, and motion-direction selectivity on naturalistic visual stimuli. Code predominantly in Python.
- Formulated a three-layer ReSU extension combining oppositely tuned T4a/T4b-like motion detectors, deriving a past–future CCA-based learning procedure for hierarchical predictive feature extraction.

Summer Research Intern; Advisor: Dmitri "Mitya" Chklovskii May 2024 - Aug. 2024

- Migrated CaImAn's deep-learning components from TensorFlow to PyTorch, including model training, inference, and pretrained-weight conversion, while validating cross-framework consistency. Code predominantly in Python.
- Reintegrated scalable calcium- and voltage-imaging analysis workflows with PyTorch backends, training and evaluating models on over 10 TB of imaging data using Flatiron's Slurm-based HPC infrastructure.
- Collaborated with HHMI Janelia researchers on developing and validating neural-imaging analysis pipelines.

Research Intern; Advisor: Jingpeng Wu and Dmitri "Mitya" Chklovskii Sept. 2022 - Aug. 2023

- Extended NeuTorch with a supervised IsoRSUNet architecture for neuron-boundary segmentation in electron-microscopy connectomes, using 2.1 billion annotated voxels from three wasp specimens and reducing Rand-based split error by up to 52% relative to baseline. Code predominantly in Python.
- Conducted architectural ablations and developed scalable HPC workflows for model training, inference, and connectome segmentation using PyTorch Connectomics and Neuroglancer.

Columbia University, Department of Physics New York, NY

Undergraduate Researcher; Advisor: Professor Szabolcs Márka Oct. 2022 - May 2023

- Developed quantum circuits for gravitational-wave signal detection using Python and Qiskit, implementing amplitude encoding and quantum-based SNR estimation.
- Evaluated quantum and classical SNR estimates using gravitational-wave data and collaborated with Heidelberg University researchers on algorithm design and validation.

PUBLICATIONS

Research Papers

1. Alexandr Andoni, **Manuel Paez**, Krish Singal, Tian Zhang. "Fast Geometric Spanners via Approximate Nearest Neighbor Search." *Innovations in Theoretical Computer Science (ITCS), 2027. Accepted.*
2. Xuehao Ding, Marco Zenari, **Manuel Paez**, Dmitri B. Chklovskii. "Asymmetric Similarity Matching: A Normative Framework for Predictive Neural Computation." *Submitted to Neural Computation*, 2026.

Workshop Publications

3. **Manuel Paez** and Chryseis Xinyi Liu. "Rank-One Spectral Normalization Accelerates Projected Feature Learning." *ICML 2026 Workshop on High-Dimensional Learning Dynamics (HiLD)*, 2026. Code.

Manuscripts in Preparation

4. **Manuel Paez,** Alexandr Andoni. "Low-Dimensional ℓ_1 Embeddings of Ising Disagreement Metrics via Functional Inequalities." Manuscript in preparation, 2026.

PRESENTATIONS

Manuel Paez and Chryseis Xinyi Liu. "Rank-One Spectral Normalization Accelerates Projected Feature Learning." Presented at the ICML 2026 Workshop on High-Dimensional Learning Dynamics, Seoul, South Korea, July 2026.

Manuel Paez. "CaImAn PyTorch: Calcium Image Data Analysis Tool for PyTorch." Simons Foundation Summer Intern Research Symposium, New York, NY, August 2024.

Manuel Paez. "Self-supervised Neuron Boundary Inpainting to Fix Membrane Leaking in Electron Microscopy Images." Poster presented at the Simons Foundation SURFiN Research Symposium, New York, New York, April 2023.

TEACHING EXPERIENCE

Columbia University New York, NY

Course Assistant, Department of Computer Science Sept. 2024 - Dec. 2025

- Organized Recitations, held office hours, and graded theoretical and programming assignments for 40+ students in COMS 4995: Randomized Algorithms and COMS 6998: Algorithms for Massive Data.

AWARDS

Dean's List — Summer 2021, Fall 2022, Spring 2024 - Spring 2025
Climate Science Olympiad 2024, Semi-finalist — Aug. 2024
MIT IQuHACK Hackathon 2023; Covalent x IBM Challenge - 1st place — Jan. 2023
Simons Foundation Global Brain SURF Fellowship — Aug. 2022
United States International Young Physicist Tournament (USIYPT) - 1st place — Feb. 2018 and Jan. 2019

ACADEMIC ACTIVITIES AND LEADERSHIP EXPERIENCE

Columbia Undergraduate Quantum Computing Club (CUQCC) — Dec. 2022 - May 2023
Co-head and Co-founder

- Co-founded and led Columbia's undergraduate quantum computing club, organizing technical seminars and research talks featuring students, faculty, and scientists in quantum information science.
- Coordinated club programming, maintained the website, and managed weekly communications with members.
- Led a team to first place in the IBM × Covalent Challenge at MIT iQuHACK 2023, developing a quantum-chaos-inspired approach to energy-grid optimization.

Markov Chains and Mixing Times Reading Group — May 2025 - Aug. 2025
Columbia Undergraduate Theoretical Computer Science Learning Seminar — Sept. 2023 - May 2026
Undergraduate Math Society (UMS) — Sept. 2019 - Dec. 2024
Supervised Undergraduate Mathematics Reading Group - Combinatorics — Sept. 2023 - Dec. 2023

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB, R, Java
Machine Learning and Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn
Scientific Computing & Data Analysis: NumPy, SciPy, pandas, MATLAB, Jupyter
Systems & High-Performance Computing: Linux, CUDA, Slurm, GPU Computing, HPC Clusters, Git
Research and Specialized Tools: TensorBoard, LaTeX, Qiskit, Cirq, Neuroglancer
Natural Languages: English (Fluent), Spanish (Fluent), German (Professional working proficiency), French (Elementary), Korean (Elementary)